

FINAL YEAR MAJOR PROJECT



Academic Year 2023-2024

Department of Computer Science and Engineering

Sree Buddha College of Engineering , Pattoor

SREE BUDDHA COLLEGE OF ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Integrated Home Automation System

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INTRODUCTION

Home automation systems, incorporating biometric authentication and facial recognition technologies, offer enhanced security and energy efficiency. This project demonstrates the integration of hardware and software components, including Microcontrollers, fingerprint sensors and various other sensors and relay circuits. The user-friendly system allows for easy configuration and management, enhancing security and personalization. This project serves as a prototype for a smart home solution, adapting to various residential and commercial settings.

PROBLEM STATEMENT

It addresses security challenges in home automation systems, highlighting the need for a user-friendly, cost-effective solution. The proposed system incorporates advanced security features like ensuring efficiency, adaptability, and security for individual user preferences.

OBJECTIVE

- Enhance Security
- Improve Safety
- Increase Convenience
- Optimize Comfort and Energy
- Centralized Control and Monitoring

SCOPE

- Integration with Voice Assistants
- Expanded Device Compatibility
- Enhanced Security Features
- Health and Wellness Monitoring

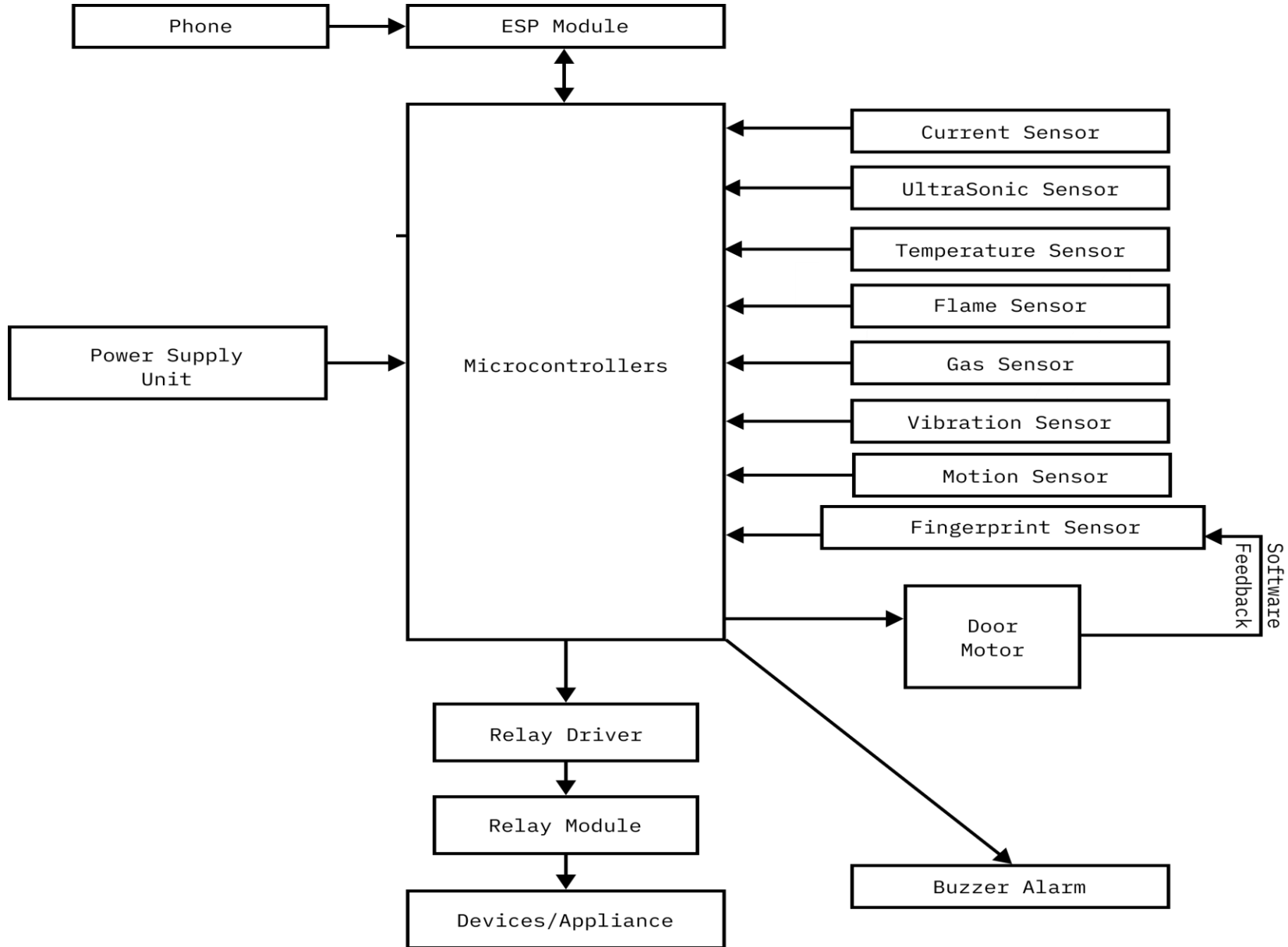
LITERATURE REVIEW

SINO	PAPER TITLE	AUTHOR NAME	METHOD USED	ADVANTAGES	DISADVANTAGES
1.	Smart Home Automation using AI and IOT with High Security	K Pushpa Rani M Ravi Kanth Pogula Sreedevi	Raspberry Pi	Increased convenience Energy efficiency User customization	Privacy concerns Complexity of setup Potential for system failures
2.	Design and Implementation of a Smart Home Automation System	ILESANMI BANJO OLUWAFEMI, TAYO DORCAS OBASANYA	Using Arduino ATMEGA328P	Enhanced Security Energy Efficiency Real-Time Alerts User-Friendly Interface	Limited Sensor Range False Alarms Limited Scalability Dependency on GSM Network
3.	AI-based Home Automation using Voice Recognition and Biometric Finger Print Authentication	Bhasha Pydala M Bharagavi Balaji Vyshnavi	Using ESP 8266 microcontroller	Enhanced security User friendly interaction Cost effective	False positives/negatives Environmental limitations Dependence on technology
4.	Integrative Home Automation System- A Multifaceted Approach	Dhyanik Pujara Riya Gautam Dhaval Shah	Using Arduino	Comprehensive integration Energy efficiency Room level Security	Cost considerations Potential security vulnerabilities Dependence on internet connectivity

PROPOSED SYSYTEM

- Gas leak sensors that send alerts to the app and shut off the gas supply automatically.
- Integrate temperature and motion sensors to automatically control the fan based on room conditions.
- Use motion sensors to automatically turn on/off lights when someone enters or leaves a room.
- Implement a biometric lock system that uses fingerprint and facial recognition for secure access.
- Utilize smoke and heat detectors that alert the app and emergency services in case of a fire.
- An intuitive app to control devices, monitor consumption, and receive real-time alerts.

ARCHITECTURE



COMPARATIVE STUDY

	EXISTING SYSTEM	PROPOSED SYSTEM
Efficiency	May have slow processes or outdated methods	Designed for faster, more efficient operations
Technology	Uses older or less advanced technology	Incorporates modern, up-to-date technology
Cost	May have high operational or maintenance costs	More cost-effective over time due to automation or optimization
User Experience	Could be complex or less user-friendly	Emphasizes ease of use and better interfaces
Security	May lack robust security features	Includes advanced and secure measures to prevent breaches

METHODOLOGY

Biometric Door Lock

Input: Fingerprint scan

Process: Fingerprint is matched against stored data

Output: Door unlocks if match is found, triggers an alarm if tampered with

Fire Extinguisher

Input: Fire detection via sensors (heat, smoke)

Process: Sensor detects fire; triggers water pump and buzzer

Output: Water is pumped to extinguish the fire, and users are alerted via app

Contactless Doorbell

Input: Ultrasonic sensor detects presence

Process: Sensor detects proximity of a person at the door

Output: Doorbell sound is triggered automatically

Gas Detector

Input: Gas concentration levels in the air

Process: Monitors for dangerous gas levels (e.g., methane, CO)

Output: Shuts off the gas regulator, activates exhaust, alarm, and sends an alert via the app

Temperature-Based Fan Speed Controller

Input: Room temperature and motion detection

Process: Temperature is monitored; fan speed adjusts based on room temperature and motion presence

Output: Fan speed changes to maintain comfort and optimize energy efficiency

Light On/Off Using Motion Sensor

Input: Motion detection in the room

Process: Motion sensor detects room occupancy

Output: Lights turn on/off depending on occupancy

Smart Blinds

Input: Ambient light levels or user command via app

Process: Light levels are sensed, or the user sets preferences via app

Output: Blinds adjust to provide optimal lighting conditions

Surveillance Cameras

Input: Real-time video feed

Process: Camera captures and streams video footage

Output: Video is either streamed live or stored for later review

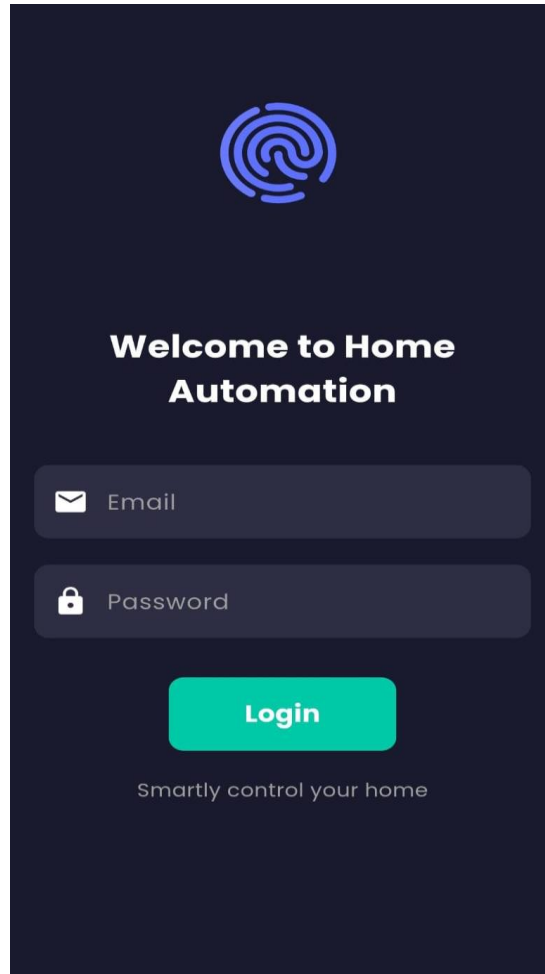
Mobile App

Input: User control or device status updates

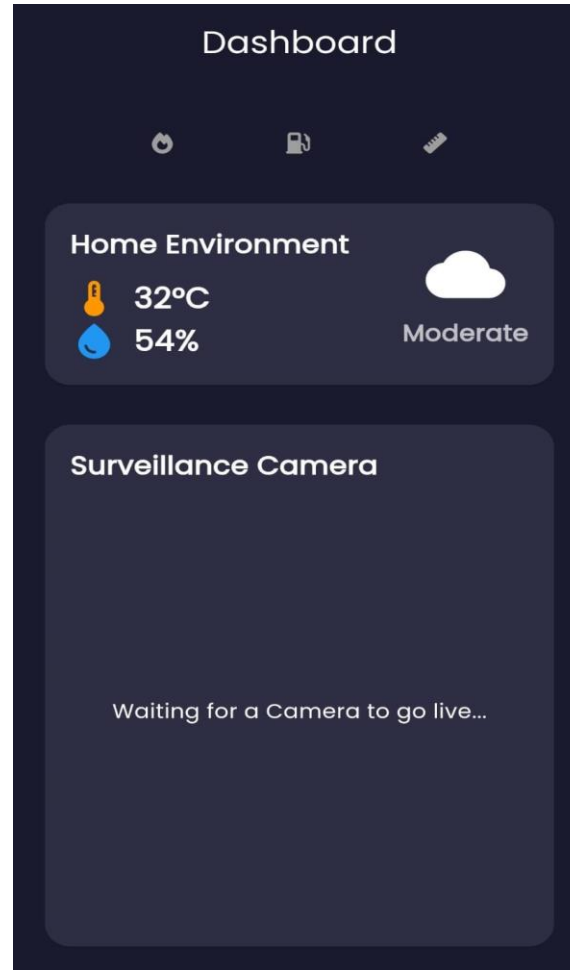
Process: App interacts with all connected devices, sending commands or receiving data

Output: Users control devices, view live feeds, check logs, and receive notifications. This methodology ensures that each device operates autonomously but can be monitored and controlled remotely for added convenience and safety.

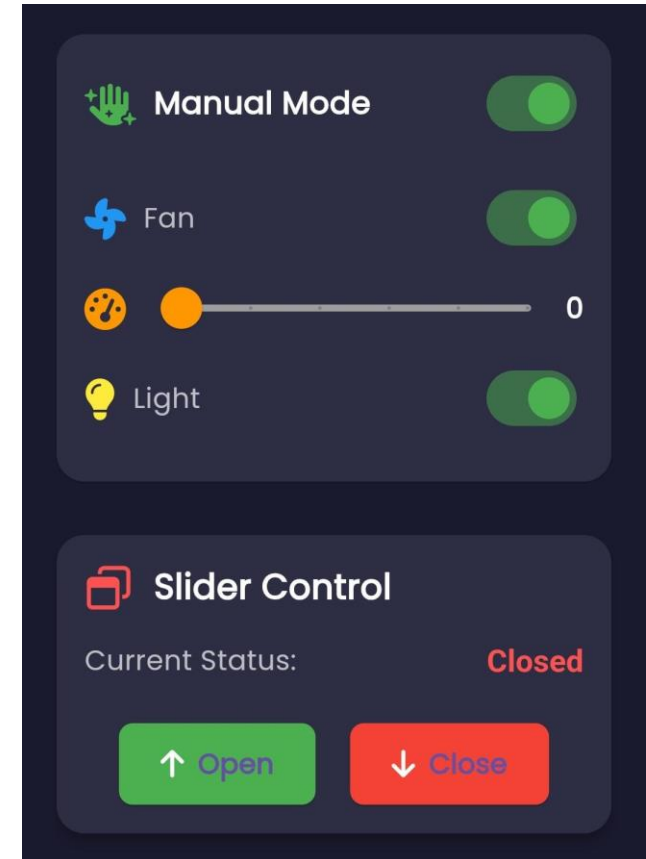
RESULT



Login Page



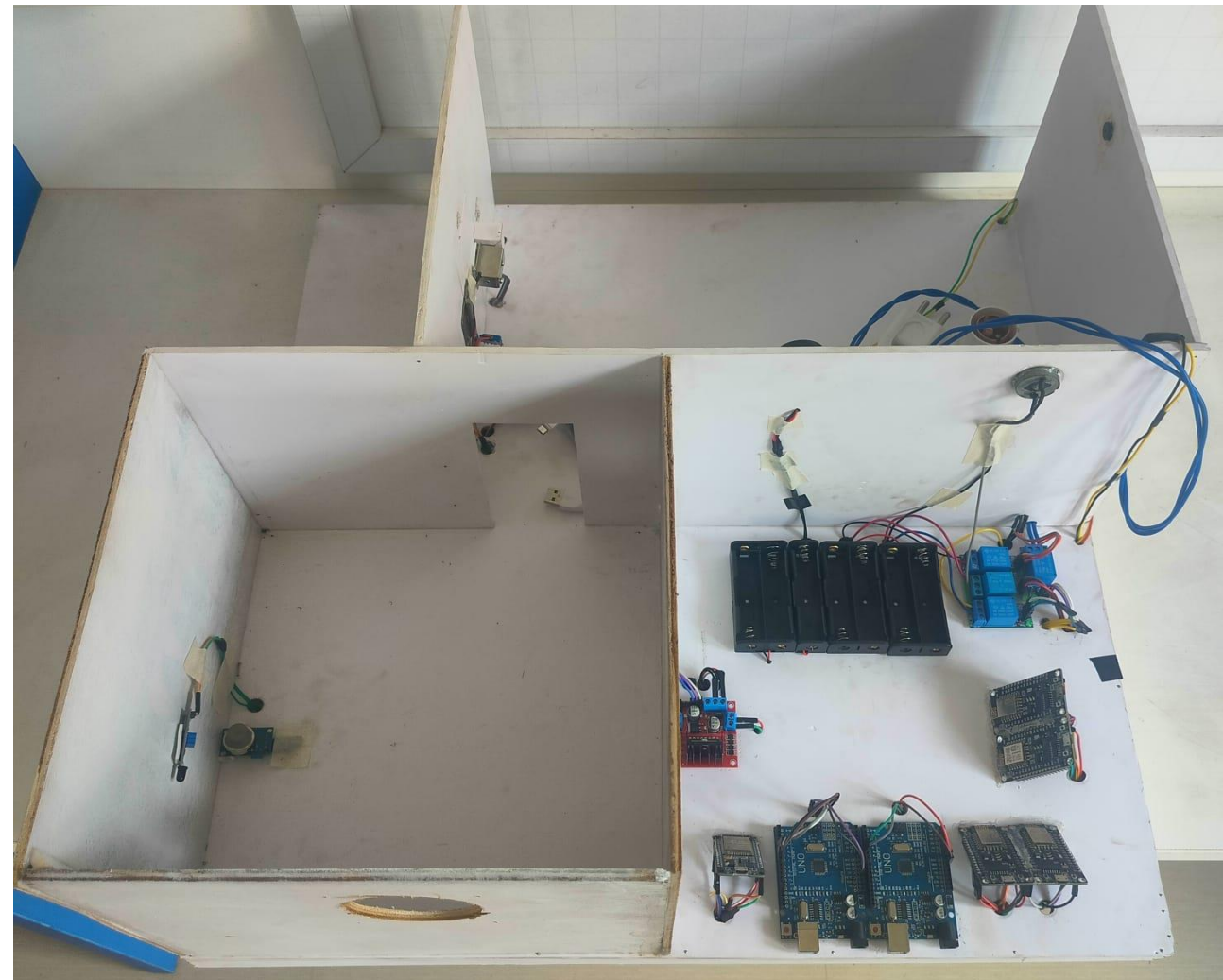
Home Page(1)



Home Page(2)

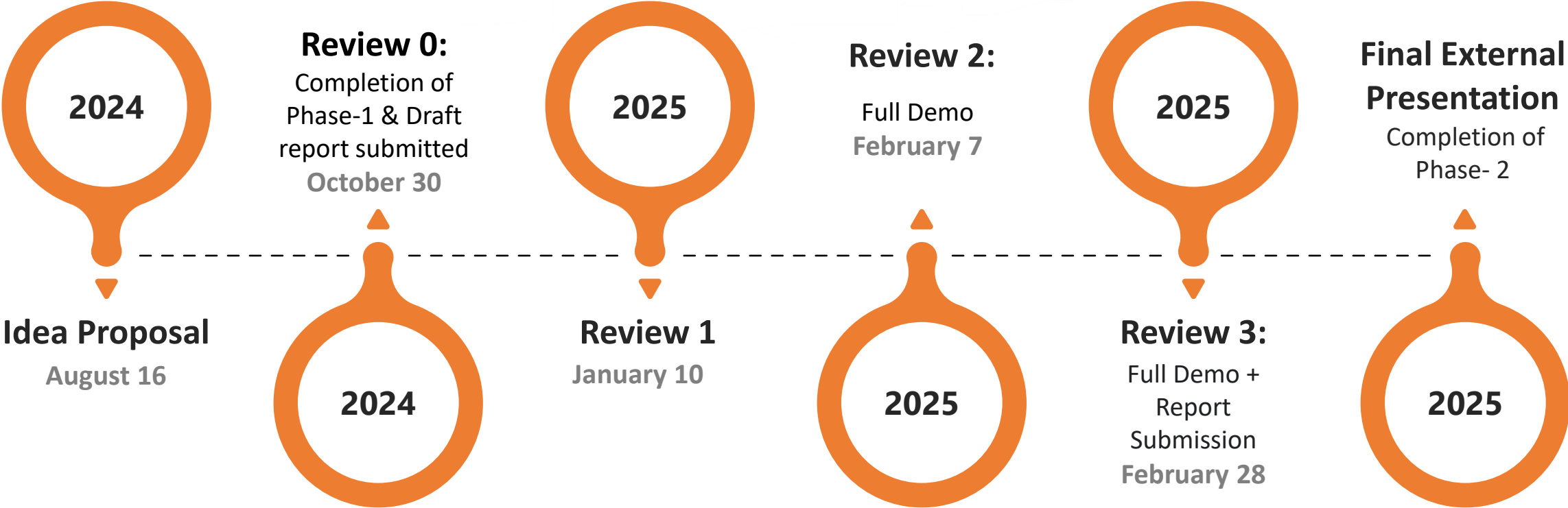


Home Page



Model

TIMELINE



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1. K. Rani, et al., "Smart Home Automation Using AI and IoT with High Security," in 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC), Bhubaneswar, India, 2024.
2. B. Pydala, et al., "AI-Based Home Automation Using Voice Recognition and Biometric Finger Print Authentication," in 2024 International Conference on Expert Clouds and Applications (ICOECA), Bengaluru, India, 2024.
3. D. Pujara, R. Gautam and D. Shah, "Integrative Home Automation System - A Multifaceted Approach," in 2023 IEEE International Symposium on Smart Electronic Systems (iSES), Ahmedabad, India, 2023.
4. H. Cao, Y. Gong and Y. Cao, "Design of Raspberry Pi-based IoT Smart Home System," in 2023 2nd International Conference on Data Analytics, Computing and Artificial Intelligence (ICDACAI), Zakopane, Poland, 2023.

Q&A

THANK YOU